Lecture 2.2 – Transient Response and Basic Stability

Week 2: Describing Performance and Checking the Motor Model

Learning Objectives

By the end of this lecture, you should be able to:

- distinguish transient response from steady-state response;
- estimate rise time and settling time;
- explain why the ideal first-order response has no overshoot;
- relate the pole $s = -1/\tau$ to response speed;
- distinguish stable and unstable behavior conceptually; and
- compare a measured motor response with a first-order model.

1. Why Describe the Response?

Knowing the final motor speed is not enough.
A controller also needs to produce acceptable behavior while the speed is changing.
For example, two motors might both reach

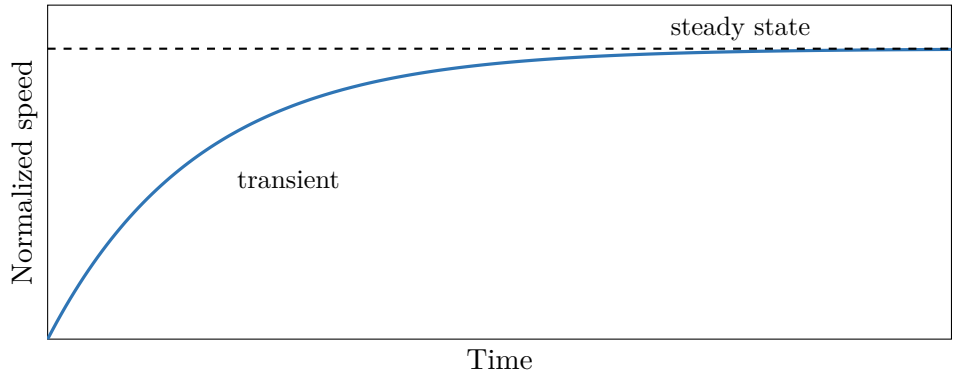
150 RPM,

but one may reach it in 0.5 s while the other takes 3 s.
The final value is the same, but the transient performance is different.

> Control engineering is concerned with both **where the response ends** and **how it gets there**.

2. Transient and Steady-State Response

The **transient response** is the part of the response while the output is changing.
The **steady-state response** is the behavior after the transient has mostly disappeared.



3. Rise Time

Rise time describes how quickly the response rises from a small fraction of its final value to a large fraction.

A common definition uses 10% to 90%.

For a first-order system,

$$\boxed{t_r \approx 2.2\tau.}$$

This relationship does not need to be memorized without meaning: it simply says that the rise time is directly related to the time constant.

Worked Example 1

If

$$\tau = 0.50 \text{ s},$$

then

$$t_r \approx 2.2(0.50) = 1.10 \text{ s}.$$

4. Settling Time

Settling time describes how long it takes the response to enter and remain close to its final value.

A common criterion is the 2% settling band.

For a first-order system,

$$\boxed{t_s \approx 4\tau.}$$

For example, if

$$\tau = 0.45 \text{ s},$$

then

$$t_s \approx 4(0.45) = 1.8 \text{ s}.$$

Smaller τ means smaller rise time and smaller settling time. The system responds faster.

5. First-Order Systems and Overshoot

The ideal first-order response is

$$y(t) = KU_0(1 - e^{-t/\tau}).$$

For positive K, positive U_0, and positive τ, the response increases smoothly toward its final value. It does not cross above the final value and then return.
Therefore,

$$\boxed{\text{an ideal first-order step response has no overshoot.}}$$

If your measured motor data appear to overshoot slightly, possible reasons include:

- measurement noise;
- encoder-count timing;
- changing load;
- supply variation;
- behavior not captured by the simple model.

6. Comparing Fast and Slow Responses

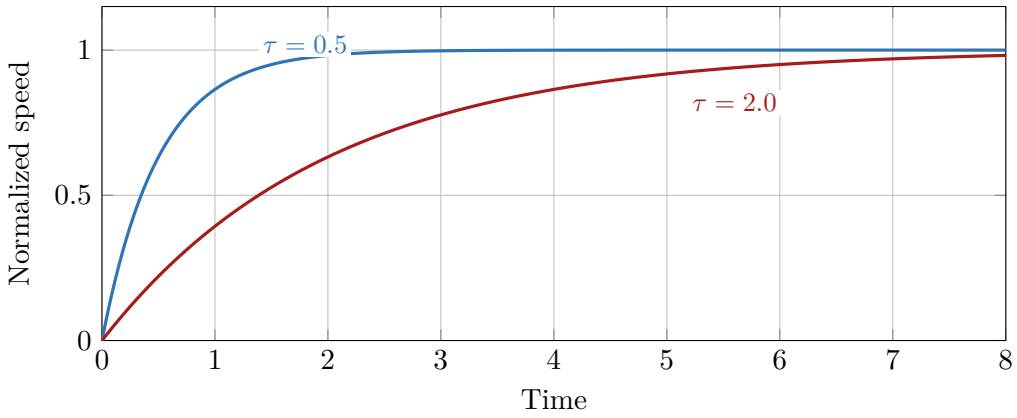
Consider two systems with the same gain K but different time constants.

$$G_1(s) = \frac{K}{0.5s + 1}, \qquad G_2(s) = \frac{K}{2s + 1}.$$

Both systems have the same steady-state gain.
However,

$$\tau_1 = 0.5 \text{ s}, \qquad \tau_2 = 2 \text{ s}.$$

Therefore, $G_1(s)$ responds faster.



7. Pole of the First-Order System

For

$$G(s) = \frac{K}{\tau s + 1},$$

the pole is found by setting the denominator equal to zero:

$$\tau s + 1 = 0.$$

Therefore,

$$\boxed{s = -\frac{1}{\tau}.}$$

Examples:

τ	Pole	Relative speed
0.5	-2	fast
1.0	-1	medium
2.0	-0.5	slow

The farther left the pole is on the real axis, the faster the first-order response.

8. Pole Location and Time Response

If

$$\tau = 0.5,$$

then

$$s = -2.$$

If

$$\tau = 2,$$

then

$$s = -0.5.$$

The pole at -2 is farther to the left and corresponds to the faster response.

> For the simple first-order systems in this course, pole location gives another way to describe response speed.

9. Basic Stability

For now, we use a simple engineering definition.

A **stable system** produces a response that remains bounded and settles to a finite value after a finite input.

An **unstable system** produces a response that grows without bound or moves farther away instead of settling.

For our physical first-order model,

$$G(s) = \frac{K}{\tau s + 1},$$

with

$$\tau > 0,$$

the pole is

$$s = -\frac{1}{\tau} < 0.$$

This produces a stable first-order response.

> This is only a basic introduction to stability. We are not using Routh-Hurwitz, root locus, Nyquist plots, or other advanced stability methods in this course.

10. Stable and Unstable Examples

A stable first-order-type response may look like

$$y(t) = 1 - e^{-t}.$$

As $t \to \infty$,

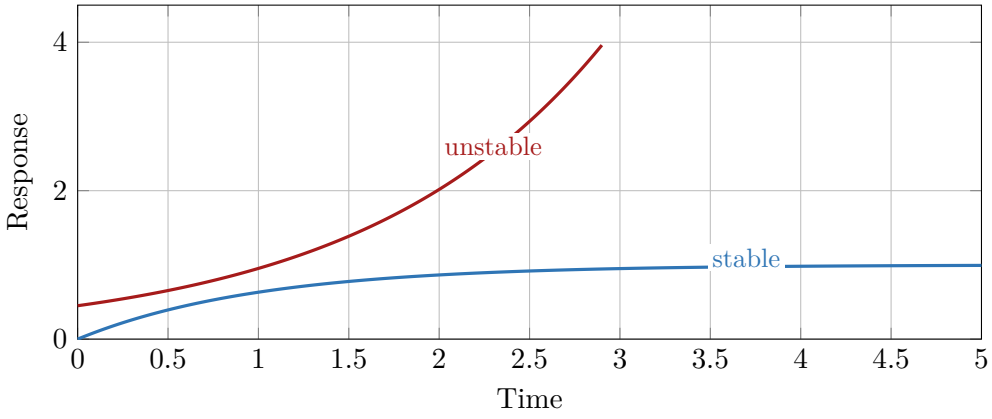
$$y(t) \to 1.$$

An unstable exponential response may look like

$$y(t) = e^{0.5t}.$$

As $t \to \infty$,

$$y(t) \to \infty.$$



11. Model Validation

After estimating K and τ, we should compare the model with the actual motor.
This is called **model validation**.
The procedure is:

1. Apply the same input to the real motor and the model.
2. Plot measured speed versus time.
3. Plot simulated speed versus time.
4. Compare:
   - final value;
   - response speed;
   - overall shape.

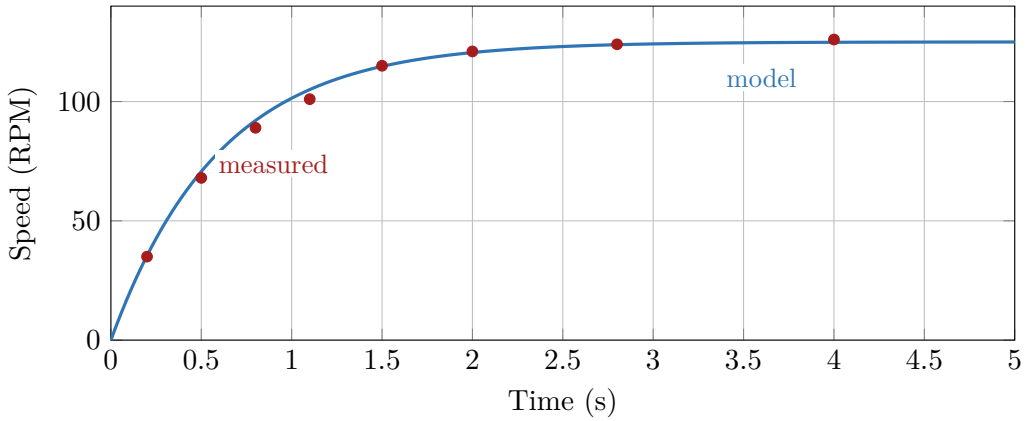
12. Example: Measured Data vs. Model

Suppose our estimated model predicts

$$y(t) = 125\left(1 - e^{-t/0.60}\right).$$

Measured data may look like:

Time (s)	Measured speed (RPM)
0.2	35
0.5	68
0.8	89
1.1	101
1.5	115
2.0	121
2.8	124
4.0	126



The model is not exact, but it captures the main trend.

13. Diagnosing Model Error

Suppose the model and measured data have the same final value, but the model rises too quickly.
Which parameter is most likely incorrect?
Since the final value is correct,

$$K$$

is probably reasonable.
The response speed is controlled mainly by

$$\tau.$$

If the model rises too quickly, its τ is too small.

Therefore, we should increase the estimated value of τ.

Rule of thumb:

- Wrong final value $\rightarrow$ check K.
- Wrong response speed $\rightarrow$ check τ.

14. Python Comparison

Measured data and model response can be plotted together.

```
import numpy as np
import matplotlib.pyplot as plt

t_meas = np.array([0.2, 0.5, 0.8, 1.1, 1.5, 2.0, 2.8, 4.0])
rpm_meas = np.array([35, 68, 89, 101, 115, 121, 124, 126])

K = 0.781
tau = 0.60
U0 = 160

t = np.linspace(0, 5, 400)
rpm_model = K * U0 * (1 - np.exp(-t / tau))

plt.plot(t, rpm_model, linewidth=2, label="Model")
plt.plot(t_meas, rpm_meas, "o", label="Measured")
plt.grid(True)
plt.xlabel("Time (s)")
plt.ylabel("Motor Speed (RPM)")
plt.legend()
plt.show()
```

15. Check Your Understanding

Answer briefly.

1. What is the difference between transient response and steady-state response?
2. If $\tau = 0.6$ s, estimate the rise time.
3. If $\tau = 0.6$ s, estimate the settling time.
4. Which pole corresponds to the faster response: -0.5 or -2?
5. Is $G(s) = K/(2s + 1)$ stable for positive K?
6. If a model reaches the correct final speed but responds too slowly, should τ be increased or decreased?

Connection to Lab 2

In Lab 2, you will use the encoder motor to:

1. apply a PWM step;
2. record RPM versus time;
3. estimate K;
4. estimate τ;
5. create
$$G(s) = \frac{K}{\tau s + 1};$$
6. simulate the model in Python; and
7. compare simulation with measured data.

Summary

- Transient response describes behavior while the output is changing.
- Steady-state response describes the long-term value.
- For a first-order system,
$$t_r \approx 2.2\tau, \qquad t_s \approx 4\tau.$$
- The ideal first-order response has no overshoot.

- The first-order pole is
$$s = -1/\tau.$$
- Negative first-order poles produce stable responses.
- Model validation compares measured response with simulated response.